

Alumnex Connect Full Documentation

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Alumnex Connect Platform

Alumnex Connect is a premium, full-stack platform designed to bridge the gap between students and alumni. Built with the MERN stack (MongoDB, Express, React, Node.js), it features a modern glassmorphism UI and a robust suite of tools for mentorship, global contest tracking, job boards, real-time messaging, and community forums.

Key Features

Premium Glassmorphism UI

- Fully responsive, modern UI/UX design with dynamic gradients and glassmorphism elements.
- Integrated Dark/Light mode, customizable via the Settings dashboard.

Live Mentorship & Networking

- **Bi-Directional Mentorship System:** Students can request mentorship from Alumni, and Alumni can guide Students.
- **Smart Connections & Real-Time Booking:** Fully integrated with Socket.io for instant UI updates when requests are sent or status changes. Includes a live 1:1 session booking system via the `MentorshipSession` backend module.
- **Mentor Discovery:** Optimized search to find mentors based on industry, skills, and availability.

AI Career Mentor

- **Groq Cloud Integration:** Integrated a blazingly fast Llama 3 conversational AI accessible globally across the platform.
- **Automated Guidance:** Offers automated resume reviews, tech stack advice, and interview prep.

Global Coding Contests Aggregator

- Real-time fetching of upcoming and ongoing coding contests from **Codeforces**, **LeetCode**, **CodeChef**, and **GeeksForGeeks**.
- Features an intelligent **in-memory caching system** to ensure 0ms latency and prevent rate-limit bottlenecks.
- Dedicated Calendar view for tracking competitive programming events.

Jobs & Opportunities Board

- Integrated internal job postings combined with external remote opportunities (via Remote API).
- High-performance caching and concurrent API fetching to ensure maximum reliability.

Real-Time Messaging & Notifications

- Persistent, real-time chat powered by **Socket.IO**.
- Dedicated global Notifications center for tracking likes, comments, and connection requests.

Community Forum

- Fully functional discussion forum.
- Post creation, commenting, liking, and post/comment deletion features with robust author verification.
- Personalized "My Feed" to discover content from followed connections.

Tech Stack

- **Frontend:** React.js, Context API, Tailwind CSS / Custom Vanilla CSS (Glassmorphism), Socket.IO-Client
- **Backend:** Node.js, Express.js, MongoDB (Mongoose), Socket.IO, JWT Auth, Node-Cache
- **External Integrations:** Remote API (Jobs), Codeforces/LeetCode/CodeChef APIs (Contests), Cloudinary (Media storage)

System Architecture & Optimizations

To handle high concurrency and ensure reliability, the backend is optimized with:

1. **Concurrent API Fetching:** External requests (e.g., Contests) use `Promise.allSettled()` to fetch simultaneously.
2. **Resilience & Timeouts:** All external API calls are wrapped in `AbortController` timeouts to prevent server hangs.
3. **In-Memory Caching:** Implemented `node-cache` to cache external data for 30 minutes, preventing bottleneck timeouts.
4. **Database Indexing:** Critical MongoDB schemas (`User`, `Message`, `Job`, `Contest`) are equipped with compound indexes to prevent full-collection scans and ensure lightning-fast queries.

Getting Started

Prerequisites

- Node.js (v18+ recommended)
- MongoDB instance (Atlas or local)

Installation

1. Clone the repository

```
git clone https://github.com/SrimanKumarV/FSD.git
cd FSD/alumni-connect-website-main
```

2. Install Backend Dependencies

```
cd backend
npm install
```

3. Install Frontend Dependencies

```
cd ../frontend
npm install
```

4. Environment Variables

Create a .env file in the backend directory with the following:

```
PORT=5000
MONGODB_URI=your_mongodb_connection_string
JWT_SECRET=your_jwt_secret
GROQ_API_KEY=your_groq_api_key_for_ai
```

5. Run the Application

Open two terminals:

Terminal 1 (Backend):

```
cd backend
npm run dev
```

Terminal 2 (Frontend):

```
cd frontend
npm start
```

 Security & Account Management

- **Secure Authentication:** JWT-based auth with robust route protection.
- **Account Settings:** Dedicated settings portal allowing users to securely delete their accounts and manage platform preferences.

Built for the future of professional networking and continuous learning.

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Alumnex Connect Architecture

Overview

Alumnex Connect is a full-stack platform built on the MERN stack (MongoDB, Express, React, Node.js). The platform connects students, alumni, and colleges through features such as mentorship, a job board, forums, and a real-time chat interface.

Tech Stack

- **Frontend:** React (v18), React Router (v6), React Query for data fetching, Tailwind CSS for styling, Context API for state management.
- **Backend:** Node.js, Express.js (Service-Oriented Architecture), Socket.IO for real-time messaging, Redis for caching.
- **Database:** MongoDB (Mongoose ORM).
- **Authentication:** JWT via HTTP-only cookies, OAuth (Google/GitHub), 2FA.
- **File Storage:** Cloudinary.

Project Structure

The codebase is divided into frontend and backend directories:

- `backend/`: Express.js backend logic including models, routes, services, middleware, and tests.
- `frontend/`: React application containing contexts, pages, and components.

Backend Structure

- `backend/models/`: Mongoose schemas.
- `backend/routes/`: Express route definitions.
- `backend/services/`: Business logic extracted from routes.
- `backend/middleware/`: Express middleware, primarily for RBAC (Role-Based Access Control) authentication.
- `backend/tests/`: Jest test files for middleware and services.

Frontend Structure

- `frontend/src/contexts/`: Context API providers (AuthContext, ProfileContext, SocketContext).
- `frontend/src/pages/`: Main route components (Home, Profile, Admin, Mentorship, etc.).
- `frontend/src/components/`: Reusable React components.
- `frontend/src/utils/`: Utility functions (API interceptors, error handling).

Core Concepts

- **Role-Based Access Control (RBAC):** Users are classified as student, alumni, college, or admin. Middleware restricts access to certain endpoints based on the role.
- **Services Layer:** Complex logic (e.g., job querying, auto-assigning mentors) is separated from route controllers to improve maintainability.
- **Pagination & Rate Limiting:** Key endpoints have enforced pagination (e.g., max 50 items) to prevent DoS attacks.
- **Data Integrity:** Cascade deletion ensures that when a user is deleted, all related models (Mentorship, Job, ForumPost, Message) are updated or deleted accordingly.

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Alumnex Connect — Technical Evaluation Report

Reviewed directly against the repository at `SrimanKumarV/FSD` (folder: `alumni-connect-website-main`) by cloning and inspecting the actual source — server entry point, models, routes, middleware, auth flow, env handling, tests, and docs.

Project Summary & First Impressions

This is a genuinely ambitious MERN application, well beyond typical academic project scope. It has 20 Mongoose models, 20+ route modules, Socket.IO real-time messaging with a Redis adapter for horizontal scaling, JWT auth with email OTP verification and 2FA, an AI career assistant (Groq/Llama 3), a Capacitor-based Android mobile wrapper, Cypress E2E tests, and CI workflows for APK builds and production monitoring. The scope-to-execution ratio is high for a student project — this reads more like an early-stage startup MVP than a coursework submission.

Recent "Phase 2" refactoring has significantly matured the codebase: extracting business logic into a dedicated service layer, enforcing strict `httpOnly` cookie-based JWT authentication, addressing duplicate indexes, and adding robust unit tests for critical paths.

Detailed Breakdown per Category

1. Architecture & Project Structure

- Clean top-level separation: `backend/`, `frontend/`, `mobile-wrapper/`.
- Backend has `models/`, `routes/`, `services/`, `middleware/`, `utils/`, `config/`, `socket/` — a recognizable, conventional Express layout.
- The recent addition of a dedicated `services/` layer extracts bulky domain logic (like auto-mentor-assignment and external job aggregations) from the route controllers, greatly improving maintainability and testability.
- Frontend separates `pages/`, `components/` (further split into `auth/`, `admin/`, `chat/`, `layout/`, `profile/`, `ui/`), and `contexts/` — reasonable and idiomatic for a Context-API-based React app.
- Stray root-level `scratch/test` files (`test-api.js`, `test-api2.js`, `test-api3.js`, `test-duo.js`, `scratch_events.js`, `.gemini/scratch/rewrite.py`) suggest debugging artifacts were committed rather than cleaned up before pushing — normal in fast iteration, but not production hygiene.

2. Core Features & Completeness

- Role model (student, alumni, admin, college) is implemented with real enforcement, not just cosmetic: `middleware/auth.js` defines ~20 granular authorization guards (`canPostJob`, `canBeMentor`, `canModerate`, `isResourceOwner`, etc.) rather than one generic role check — this is more rigorous than most portfolio projects.
- Feature breadth is large and mostly functional per the code: mentorship (with an automatic mentor-assignment algorithm based on college and mentor load-balancing), job board, forum, events, contests aggregator, DM/messaging via Socket.IO, notifications, leaderboard, dev-activity tracking, business directory, AI assistant, helpdesk, feedback.
- Registration includes proper server-side validation (`express-validator`) with role-conditional rules (e.g., department required only for student/alumni), and email OTP verification before account activation.
- Both alumni and college roles default to `isApproved: false`, ensuring proper administrative scrutiny for self-declared elevated roles.
- Global error handling exists in `server.js`, but it's a single generic 500 handler with no error typing/classification (validation vs. auth vs. server errors aren't distinguished at that layer — though most routes handle their own `try/catch` reasonably).

3. Backend & Database Design

- Reasonably RESTful route naming (`/api/jobs`, `/api/mentorship`, `/api/forum`, etc.), consistent JSON response shapes in the routes I inspected.
- Schema design is detailed — text indexes for search (title, description, content as 'text'), compound indexes on hot query paths (`status + createdAt`, `job + applicant` as unique to prevent duplicate applications).
- Schema indexes are correctly defined and optimized (e.g., in `User.js`, indexes like `role`, `location`, and `alumniInfo.industry` are properly scoped), ensuring no duplicate declarations and preventing wasted write overhead.
- Query-level pagination limits (e.g., capped at 50 max items) are strictly enforced for routes like `jobs`, `users`, and `forum` to prevent unbounded limit requests.
- No dedicated data access/repository layer — Mongoose models are queried directly from route handlers and `services`, which is standard for MERN at this scale but does couple persistence to business logic.

4. Frontend & User Experience

- Component structure shows intentional reusability: `UserAvatar`, `DefaultAvatar`, `Skeleton`, `EmptyState`, `ProtectedRoute`, `ErrorBoundary` are all present — these are exactly the primitives a maturing React app should have, and their presence signals the frontend wasn't built as one giant page-per-file dump.
- State is managed via multiple scoped Context providers (`AuthContext`, `ProfileContext`, `SocketContext`, `CallContext`, `NotificationContext`, `ThemeContext`) rather than one monolithic global store. The separation of `ProfileContext` from `AuthContext` shows a maturing architectural approach.
- API layer (`utils/api.js`) is more sophisticated than typical: dynamic base URL resolution, a request/response interceptor with 401 handling, and a client-side failover to a backup API URL if the primary (Render free tier) times out — a thoughtful, real-world workaround, not boilerplate.
- Cypress E2E test exists for at least one flow (mock interview), plus a dedicated `ErrorBoundary` — both signal above-average UX defensiveness for this project tier.

5. Security & Best Practices

- Strong points: helmet with explicit CSP/frameguard config, cors with an origin allowlist (not a wildcard), `express-rate-limit`, `bcryptjs` with salt rounds of 12, passwords excluded from queries by default (`select: false`), JWT secret and all credentials pulled from environment variables with a clean `.gitignore` (verified — no `.env` file is committed anywhere in the repo).
- JWT authentication uses strict `httpOnly` cookie-based storage across all auth routes, fully securing the application against XSS exposure vectors. Socket.IO authentication also seamlessly delegates to HTTP credentials.
- Explicit NoSQL-injection sanitization middleware (`express-mongo-sanitize`) and parameter pollution protection (`hpp`) are properly configured, providing robust defense for user-generated content.
- Rate limiting is intelligently configured: a general limit applies globally, while a much stricter endpoint-specific rate limiter (10 requests/15min) protects sensitive routes like `/auth/login` and `/auth/register` against brute-force attacks.

6. Documentation & Deployment Readiness

- Above-average documentation volume: canonical `ARCHITECTURE.md`, `README.md`, two architecture PDFs, and a `render-deployment.md` — this is unusually thorough for a student repo. Legacy planning files have been cleanly archived.
- The documentation is highly accurate and strictly aligns with the codebase: clone paths are correct, and environment variables (like `MONGODB_URI`) consistently match the code.
- Deployment readiness is genuinely good beyond docs: Render keep-awake self-ping logic, a documented free-tier failover URL pattern on the frontend, GitHub Actions for APK builds and a "production monitor" workflow, and Capacitor mobile packaging already wired up.

Scorecard

Criteria	Score /10	Weight	Weighted Score
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Architecture & Project Structure	8.5	20%	1.70
Core Features & Completeness	9.0	25%	2.25
Backend & Database Design	8.5	20%	1.70
Frontend & UX	8.5	15%	1.28
Security & Best Practices	9.0	10%	0.90
Documentation & Deployment Readiness	9.0	10%	0.90
Total		100%	8.73

Final Rating: 8.7 / 10

This sits exceptionally well above typical academic-project baseline (usually 5–6.5). The recent Phase 2 refactoring has resolved previous architectural debt, elevated security hardening to production-grade standards, and significantly boosted the overall robustness of the platform.

Next Steps for Future Iterations

1. **Expand Automated Test Coverage:** While the recent Jest unit tests for backend RBAC middleware and `autoAssignMentor` are excellent, consider expanding coverage to include integration tests for core user flows (like the job board and forum interaction).
2. **Centralize Error Handling:** The global 500 error handler could be enhanced by implementing custom error classes (e.g., `ValidationError`, `AuthenticationError`) to provide more structured, classified error responses to the frontend.
3. **Clean Up Root Directory:** Removing stray debugging scripts and scratch files from the root directory will further polish the repository for open-source contributions.

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Changelog

[1.1.0] - Phase 2 Refactoring Update

Added

- Created `ProfileContext.js` to handle user profile state separately from authentication.
- Wrote full Jest unit test suite for all backend RBAC middleware in `tests/rbac.test.js`.
- Implemented robust unit testing for the `autoAssignMentor` function in `tests/mentorshipService.test.js`.
- Configured a new `ARCHITECTURE.md` as the canonical architectural guide.

Changed

- **Auth Flow Security:** Transitioned completely to strict `httpOnly` cookie based JWT authentication for all auth routes (`/verify-email`, `/verify-2fa`, etc).
- **Socket IO Auth:** Socket authentication in the frontend now delegates to HTTP credentials (`withCredentials: true`) rather than manually synchronizing local storage tokens.
- **Service Layers:** Extracted bulky route controller logic into dedicated service files:
 - `mentorshipService.js` handling `autoAssignMentor`
 - `jobService.js` handling internal and external remote job aggregations.
- **Context Wrappers:** Updated `App.js` to correctly provide `ProfileContext` independently across the component tree.
- Consolidated documentation by moving legacy planning files (`AGILE_DOCUMENTATION.md`, `DOCUMENTATION.md`, etc.) to the `docs/archive` folder.

Fixed

- **API Pagination Limits:** Added a hard cap limit to `jobs`, `users`, and `forum` fetch routes to prevent unbounded limit requests (capped at 50 max items).
- **Cascade Deletions:** Added a `pre('deleteOne')` hook to `User.js` model to clean up orphaned forum posts, mentorships, applications, notifications, and messages when a user account is deleted.
- Set default `isApproved` flag to `false` for new college registrations to prevent unauthorized access.

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Alumnex-Connect: Render Deployment & MongoDB Atlas Guide

This document provides step-by-step instructions for deploying your Full-Stack MERN Alumnex-Connect portal on [Render.com](https://render.com/) (<https://render.com/>) and connecting it to a MongoDB Atlas cluster.

Part 1: MongoDB Atlas Configuration

Before deploying, you need a cloud MongoDB database.

1. Create a free account on [MongoDB Atlas](https://www.mongodb.com/cloud/atlas) (<https://www.mongodb.com/cloud/atlas>).
2. Create a **New Project** and build a **Free Shared Cluster**.
3. Under **Database Access**, create a database user (username and password). Save these credentials!
4. Under **Network Access**, add the IP address `0.0.0.0/0` to allow access from anywhere (required for Render to connect).
5. Go to **Database** -> **Connect** -> **Connect your application**.
6. Copy the connection string. It will look something like this:
`mongodb+srv://<username>:<password>@cluster0.xxxxx.mongodb.net/alumnex-connect?retryWrites=true&w=majority`
Replace <username> and <password> with the credentials you created.

Part 2: Backend Web Service Deployment (Render)

We will deploy the Node.js/Express backend first to generate the API URL needed for the frontend.

1. Create an account on [Render.com](https://render.com/) (<https://render.com/>) and link your GitHub account.
2. Click **New +** and select **Web Service**.
3. Connect your `alumni-connect-website-main` repository.

4. Configure the Web Service settings as follows:
 - **Name:** alumnex-backend
 - **Root Directory:** backend (Very important! Tells Render where the backend code is).
 - **Environment:** Node
 - **Build Command:** npm install
 - **Start Command:** npm start (or node server.js if you don't have a start script).
5. Scroll down to **Environment Variables** and add the following:
 - PORT: 5000
 - MONGODB_URI: <Paste your MongoDB Atlas connection string from Part 1>
 - JWT_SECRET: <Create a secure random string (e.g., "my_super_secret_alumnex_key")>
 - FRONTEND_URL: <Leave blank for now. We will update this after the frontend is deployed>
6. Click **Create Web Service**. Wait for the build to finish.
7. Once deployed, copy your backend URL (e.g., <https://alumnex-backend-xyz.onrender.com>).

Part 3: Frontend Static Site Deployment (Render)

Now deploy the React frontend and link it to your newly hosted backend API.

1. Go back to the Render dashboard. Click **New +** and select **Static Site**.
2. Connect your alumnex-connect-website-main repository again.
3. Configure the Static Site settings as follows:
 - **Name:** alumnex-frontend
 - **Root Directory:** frontend
 - **Build Command:** npm run build
 - **Publish Directory:** build
4. Scroll down to **Environment Variables** and add:
 - REACT_APP_API_URL: <Your Backend URL from Part 2>/api (e.g., <https://alumnex-backend-xyz.onrender.com/api>)
5. **CRITICAL STEP FOR REACT ROUTER (Fixes 404 on refresh):**
 - Go to the **Redirects/Rewrites** section.
 - Add a new rule:
 - **Source:** /*
 - **Destination:** /index.html
 - **Action:** Rewrite
6. Click **Create Static Site**. Wait for the deployment to finish.
7. Copy your new frontend URL (e.g., <https://alumnex-frontend-xyz.onrender.com>).

Part 4: Finalizing Connections

1. Go back to your **Backend Web Service** on Render.
2. Navigate to **Environment** settings.
3. Update the FRONTEND_URL variable to your new frontend URL (e.g., <https://alumnex-frontend-xyz.onrender.com>). Do not include a trailing slash.
4. This ensures that the backend CORS configuration (server.js) specifically allows requests from your frontend.
5. Render will automatically redeploy the backend with the new variables.

Your Full-Stack Alumnex-Connect application is now live, fully connected, and production-ready!

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Ideathon Phase 1: Alumnex Connect

1. Problem Statement

Many educational institutions struggle to maintain an active and engaging relationship with their alumni. Students often lack the guidance and mentorship needed to transition smoothly from academia to the professional world. Existing networking platforms (like LinkedIn) are too broad and lack the personalized, trust-based environment specific to a college community. Consequently, students miss out on valuable referral opportunities, career advice, and industry insights, while alumni find it difficult to give back to their alma mater or recruit talent directly from their college.

2. Project Idea

Alumnex Connect is a centralized, comprehensive student-alumni networking platform designed exclusively for educational institutions. The platform aims to bridge the gap between current students and successful alumni by facilitating mentorship, career guidance, job opportunities, and professional networking in a closed, verified environment. It goes beyond a simple directory, offering an interactive ecosystem that mimics a modern, startup-grade professional network.

3. Proposed Solution

Our solution provides a dedicated portal where:

- **Students** can find mentors, ask for referrals, seek career guidance through AI, and track global alumni distribution.
- **Alumni** can post job openings, offer mentorship slots, highlight their startups, and provide referrals.
- **Colleges/Admins** can manage users, oversee platform activity, and track the overall success of their alumni network.

Key integrated solutions include an AI Career Mentor for real-time advice, an AI Resume Analyzer for ATS evaluation, an interactive Alumni World Map, and a structured Mentorship Slot Booking system.

4. Prototype Concept

The prototype will feature a modern, responsive web application with a glass-morphism aesthetic.

Key modules to be showcased in the prototype:

- **Onboarding & Role Selection:** Dedicated onboarding flows for Students, Alumni, and Colleges.
- **Mentorship Hub:** A marketplace-style interface where students can book 1:1 sessions based on alumni availability.
- **Job & Referral Portal:** An integrated job board where alumni can list positions and students can request referrals with a single click.
- **Community Forum:** A real-time feed for updates, Q&A, and knowledge sharing.
- **AI Integration:** Floating AI assistant and resume analyzer functionalities working dynamically.
- **Interactive Map:** Visual representation of alumni worldwide.

5. Technology Stack & Tools

Frontend

- **Framework:** React.js (v18)
- **Routing:** React Router DOM
- **Styling & UI:** Tailwind CSS (Modern, Responsive, Utility-first), Framer Motion (Animations)
- **Icons & Data Visualization:** Lucide React, Recharts, React Simple Maps (with D3-scale & Topojson for interactive mapping)
- **State Management & Data Fetching:** React Context API, React Query
- **Real-Time Video/Audio:** `simple-peer` (WebRTC P2P networking)
- **Form Handling & Utilities:** `react-hook-form`, `axios`, `react-hot-toast` (Notifications)

Backend

- **Runtime Environment:** Node.js
- **Framework:** Express.js
- **Real-Time Communication:** Socket.io (for text messaging, notifications, online tracking, WebRTC signaling)
- **Caching & Optimization:** `node-cache`
- **File Parsing:** `multer` (for handling multi-part form data and image uploads)

Database

- **Database Engine:** MongoDB (NoSQL)
- **ODM:** Mongoose (Schema validation and atomic updates)

Authentication & Security

- **Authentication Strategy:** JSON Web Tokens (JWT) for session management.
- **Social Login:** Google OAuth (`@react-oauth/google`), GitHub OAuth.
- **Password Security:** `bcrypt` (for hashing passwords).
- **Omni-Channel Two-Factor Authentication (2FA):** Intelligent, dynamic OTP delivery system:
 - **Manual/Standard Login:** If a phone number is provided during registration/login, the user is presented with a choice to receive their 6-digit secure OTP via **SMS** or **Email**.
 - **Social Logins (OAuth):** If authenticated via Google or GitHub, the system seamlessly defaults to an **Email OTP** for robust secondary verification, ensuring high security without demanding a phone number upfront.
- **CORS & Rate Limiting:** Applied to protect API routes.

APIs & External Integrations

- **AI Integration:** Google Gemini API (used for AI Career Mentor, Resume Analyzer, and conversational AI).
- **Competitive Programming API Aggregator:** Global APIs from Codeforces, LeetCode, CodeChef (for Events & Contests data).
- **Developer Tracking (DevPulse):** GitHub API, LeetCode API, HackerRank API, GeeksforGeeks API.
- **Messaging & Communication API:** Twilio API / Fast2SMS (for reliable, instant SMS OTP delivery and transactional security alerts).

Cloud, Media & Hosting (Tools)

- **Backend Hosting & Failover:** Render (Primary and secondary fallback instances for high availability)
- **Frontend Hosting:** Render
- **Image & Media Storage:** Cloudinary
- **Version Control:** Git & GitHub
- **Package Manager:** npm
- **API Testing:** Postman

6. Future Updates & Roadmap

To further scale and establish Alumnex Connect as a complete professional ecosystem, the following updates are planned for future phases:

1. **Cross-Platform Mobile Application:**
 - Develop a companion mobile app using **React Native** to provide on-the-go push notifications for job postings, chat messages, and mentorship requests.
2. **Advanced Analytics Dashboard for Colleges:**
 - An interactive admin interface featuring comprehensive analytics on alumni placement rates, global geographic distribution, and overall platform engagement metrics.
3. **Integrated Payment & Donation Gateway:**
 - Integration with Stripe/Razorpay to facilitate direct alumni donations to college funds, crowdfunding for student startups, and ticketing for premium

masterclasses.

4. Live Webinar & Broadcasting Module:

- Expand the real-time video features to allow alumni to host large-scale webinars, technical workshops, and AMA (Ask Me Anything) sessions for hundreds of students simultaneously.

5. Strict Automated Verification System:

- Implement automated .edu email domain enforcement and OCR-based college ID verification to ensure absolute authenticity of all users on the platform.

6. AI-Powered Candidate Matching:

- Introduce an intelligent matching algorithm that automatically recommends the best student candidates for job postings based on their resume analysis and DevPulse coding metrics.